

Problem Set 1

MTSM

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Exercise 1

Point i)

Consider a matrix polynomial defined as:

$$A(z) = I_p - \sum_{i=1}^k A_i z^i, \quad \text{with } A_i \in \mathbb{R}^{p \times p}, \quad z \in \mathbb{C},$$

and let $z_1, z_2, \dots, z_n \in \mathbb{C}$ be the characteristic roots of $A(z)$.

Show that $\frac{1}{z_i}$ is an eigenvalue of the companion matrix

$$\mathcal{A}_c = \begin{bmatrix} A_1 & A_2 & \cdots & A_{k-1} & A_k \\ I_p & 0 & \cdots & 0 & 0 \\ 0 & I_p & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & I_p & 0 \end{bmatrix}.$$

Proof

The characteristic roots z_i of the matrix polynomial are defined as the values of $z \in \mathbb{C}$ for which:

$$\det(A(z)) = \det \left(I_p - \sum_{i=1}^k A_i z^i \right) = 0.$$

This implies that $A(z)$ is singular, so there exists a non-zero vector $w \in \mathbb{C}^p$ such that:

$$A(z)w = 0,$$

i.e., $w \in \ker(A(z))$. Expanding:

$$\left(I_p - \sum_{i=1}^k A_i z^i\right) w = 0 \quad \Rightarrow \quad w = \sum_{i=1}^k z^i A_i w. \quad (1)$$

Consider the VAR(k) model:

$$z_t = \sum_{i=1}^k A_i z_{t-i} + u_t,$$

where $z_t \in \mathbb{R}^p$. Define:

$$z_t^c = \begin{bmatrix} z_t \\ z_{t-1} \\ \vdots \\ z_{t-k+1} \end{bmatrix}, \quad z_{t-1}^c = \begin{bmatrix} z_{t-1} \\ z_{t-2} \\ \vdots \\ z_{t-k} \end{bmatrix}, \quad u_t^c = \begin{bmatrix} u_t \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \text{all in } \mathbb{R}^{kp}.$$

Then:

$$z_t^c = \mathcal{A}_c z_{t-1}^c + u_t^c.$$

Let $\mathcal{A}_c v = \lambda v$, and partition $v \in \mathbb{C}^{kp}$ as:

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_k \end{bmatrix}, \quad v_i \in \mathbb{C}^p.$$

Then:

$$\begin{cases} A_1 v_1 + A_2 v_2 + \cdots + A_k v_k = \lambda v_1, \\ v_1 = \lambda v_2, \\ v_2 = \lambda v_3, \\ \vdots \\ v_{k-1} = \lambda v_k. \end{cases} \quad (1)$$

Substituting recursively, we find:

$$A_1 \lambda^{k-1} v_k + A_2 \lambda^{k-2} v_k + \cdots + A_k v_k = \lambda^k v_k.$$

Assuming $\lambda \neq 0$, divide by λ^k :

$$\frac{1}{\lambda} A_1 v_k + \frac{1}{\lambda^2} A_2 v_k + \cdots + \frac{1}{\lambda^k} A_k v_k = v_k. \quad (2)$$

But from Equation (1), any characteristic root z satisfies:

$$w = z A_1 w + z^2 A_2 w + \cdots + z^k A_k w.$$

Comparing with Equation (2), set $w = v_k$, and identify $z = \frac{1}{\lambda}$.

Conclusion

The characteristic roots z of the matrix polynomial $A(z)$ are the reciprocals of the eigenvalues λ of the companion matrix $\mathcal{A}_c$.

Point ii)

State the stability condition of a VAR(k) process in terms of the characteristic roots of its matrix polynomial and relate it to the eigenvalues of the companion matrix.

Proof

The stability of a VAR(k) process is governed by the characteristic roots of the associated matrix polynomial:

$$A(z) = I_p - \sum_{i=1}^k A_i z^i.$$

The process is said to be **stable** if all the roots $z \in \mathbb{C}$ of the characteristic equation

$$\det \left(I_p - \sum_{i=1}^k A_i z^i \right) = 0$$

lie **strictly outside the unit circle**, that is,

$$|z| > 1 \quad \text{for all roots } z.$$

This ensures that the influence of initial conditions decays over time, avoiding non-stationary or explosive behavior.

From the previous result, the characteristic roots z of the matrix polynomial are the reciprocals of the eigenvalues λ of the companion matrix $\mathcal{A}_c$, i.e.,

$$\lambda = \frac{1}{z}.$$

Therefore, the stability condition $|z| > 1$ is equivalent to:

$$|\lambda| < 1.$$

Conclusion

The VAR(k) process is stable if and only if all the eigenvalues of the companion matrix $\mathcal{A}_c$ lie strictly inside the unit circle in the complex plane.

Point iii)

Show that 0 is an eigenvalue of the companion matrix $\mathcal{A}_c$ if and only if A_k is singular.

Proof

($\Rightarrow$) **Suppose that 0 is an eigenvalue of $\mathcal{A}_c$.**

Then there exists a non-zero vector $v \in \mathbb{C}^{kp}$ such that:

$$\mathcal{A}_c v = 0.$$

Partition the vector as:

$$v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_k \end{bmatrix}, \quad v_i \in \mathbb{C}^p.$$

Then:

$$\mathcal{A}_c v = \begin{bmatrix} A_1 v_1 + A_2 v_2 + \cdots + A_k v_k \\ v_1 \\ v_2 \\ \vdots \\ v_{k-1} \end{bmatrix} = 0.$$

Which gives the system:

$$\begin{cases} v_1 = 0, \\ v_2 = 0, \\ \vdots \\ v_{k-1} = 0, \\ A_1 v_1 + A_2 v_2 + \cdots + A_k v_k = 0. \end{cases} \quad (2)$$

Substituting the first $k - 1$ equations into the last yields:

$$A_k v_k = 0.$$

Since $v_k \neq 0$, this implies A_k is singular.

($\Leftarrow$) **Now suppose A_k is singular.**

Then there exists $w \in \mathbb{C}^p$, $w \neq 0$, such that:

$$A_k w = 0.$$

Define:

$$v = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ w \end{bmatrix} \in \mathbb{C}^{kp}.$$

Then:

$$\mathcal{A}_c v = \begin{bmatrix} A_k w \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} = 0.$$

Thus, $v \neq 0$ and $\mathcal{A}_c v = 0$, so 0 is an eigenvalue of $\mathcal{A}_c$.

Conclusion

The companion matrix $\mathcal{A}_c$ has 0 as an eigenvalue if and only if A_k is singular.

Point iv)

Show that the number of characteristic roots of the matrix polynomial $A(z)$ is exactly kp if and only if A_k is nonsingular.

Proof

We define the characteristic roots as the complex values $z \in \mathbb{C}$ for which:

$$\det(A(z)) = \det \left(I_p - \sum_{i=1}^k A_i z^i \right) = 0.$$

This determinant is a scalar polynomial in z , whose degree determines the number of characteristic roots by the Fundamental Theorem of Algebra.

We now expand the determinant using the **Leibniz formula**:

$$\det(M) = \sum_{\sigma \in S_p} \text{sgn}(\sigma) \prod_{i=1}^p M_{i, \sigma(i)},$$

where:

- S_p is the symmetric group on p elements,
- $\text{sgn}(\sigma)$ is the sign of the permutation,
- $M_{i, \sigma(i)}$ denotes the $(i, \sigma(i))$ -th entry of matrix M .

In our case, the matrix $M = A(z)$, whose entries are:

$$M_{i,j} = \delta_{ij} - \sum_{\ell=1}^k (A_\ell)_{ij} z^\ell,$$

where δ_{ij} is the Kronecker delta.

We isolate the term of highest degree in $\det(A(z))$. This occurs when, for each $i = 1, \dots, p$, we select the term $-(A_k)_{i, \sigma(i)} z^k$, leading to:

$$\prod_{i=1}^p (-(A_k)_{i, \sigma(i)} z^k) = (-1)^p z^{kp} \prod_{i=1}^p (A_k)_{i, \sigma(i)}.$$

Summing over all permutations:

$$\sum_{\sigma \in S_p} \text{sgn}(\sigma) \prod_{i=1}^p (-(A_k)_{i, \sigma(i)} z^k) = (-1)^p z^{kp} \sum_{\sigma \in S_p} \text{sgn}(\sigma) \prod_{i=1}^p (A_k)_{i, \sigma(i)}.$$

This gives:

$$(-1)^p \det(A_k) z^{kp}.$$

Conclusion

- If A_k is nonsingular, then $\det(A_k) \neq 0$, so the highest-degree term in $\det(A(z))$ is of degree kp .
- If A_k is singular, then $\det(A_k) = 0$, and all terms have degree less than kp .

Exercise 2

Consider a matrix polynomial:

$$A(z) = I_p - \sum_{i=1}^k A_i z^i, \quad A_i \in \mathbb{R}^{p \times p}, \quad z \in \mathbb{C}$$

with characteristic roots z_i , for $i = 1, 2, \dots, n$.

Point i)

Show that $z_i \neq 0$ for all characteristic roots of $A(z)$.

Proof

We want to prove that zero cannot be a characteristic root of the matrix polynomial $A(z)$. Before starting, let us recall the definition of "characteristic root":

Definition: A complex number $z \in \mathbb{C}$ is called a **characteristic root** of the matrix polynomial $A(z)$ if it satisfies:

$$\det(A(z)) = 0$$

That is, the characteristic roots are the values of z for which the matrix $A(z)$ becomes singular (i.e., non-invertible).

We proceed by contradiction. Suppose that $z_i = 0$ is a characteristic root. Then, by definition, we would have:

$$\det(A(0)) = 0$$

Let us compute $A(0)$ explicitly:

$$A(z) = I_p - \sum_{i=1}^k A_i z^i \quad \Rightarrow \quad A(0) = I_p - \sum_{i=1}^k A_i \cdot 0^i = I_p - \overbrace{\sum_{i=1}^k A_i \cdot 0^i}^{=0} = I_p$$

Therefore:

$$\det(A(0)) = \det(I_p) = 1 \neq 0$$

Conclusion

This contradicts the assumption that $\det(A(0)) = 0$. Hence, $z_i = 0$ cannot be a characteristic root of the polynomial $A(z)$.

Point ii–iii)

Let $A(z)^{-1} = \sum_{i=0}^{\infty} C_i z^i$ be its inverse. Express the radius of convergence of this power series in terms of the characteristic roots.

Proof

To study the inverse of the matrix polynomial $A(z)$, we begin by considering the scalar case, which is helpful to build intuition.

Let $a(z) = 1 - \sum_{i=1}^k a_i z^i$, where $a_i \in \mathbb{R}$. In this case, the inverse is simply:

$$a(z)^{-1} = \frac{1}{a(z)} = \frac{1}{1 - \sum_{i=1}^k a_i z^i}$$

If we focus on the simpler expression $a(z) = 1 - cz$, with $c \in \mathbb{R}$, we can expand the inverse as a geometric series:

$$\frac{1}{1 - cz} = \sum_{n=0}^{\infty} (cz)^n \quad \text{for } |cz| < 1$$

This observation shows that the inverse can be represented as a power series, provided that the value of z is small enough so as not to approach any root of the denominator. In other words, the convergence of the series depends on the distance between z and the characteristic roots of the polynomial.

Let us now move to the matrix case. The inverse of $A(z)$ is not given by simple division, but rather by the standard formula for the inverse of a square matrix:

$$A(z)^{-1} = \frac{\text{adj}(A(z))}{\det(A(z))}$$

Here, $\text{adj}(A(z))$ is the adjugate matrix, that is, the transpose of the cofactor matrix of $A(z)$. Technically, each cofactor is the determinant of a submatrix obtained by removing one row and one column from $A(z)$, multiplied by the appropriate sign $(-1)^{i+j}$. Informally, the adjugate serves as the algebraic component that allows us to compute the inverse in the absence of a division operation for matrices.

This formula is valid only when $\det(A(z)) \neq 0$, meaning that $A(z)$ is invertible.

Moreover, this expression shows that the behavior of $A(z)^{-1}$ is governed by the analytic behavior of $\det(A(z))$: the inverse becomes singular precisely when $\det(A(z)) = 0$, that is, when z approaches a characteristic root.

Therefore, in the series expansion $A(z)^{-1} = \sum_{n=0}^{\infty} C_n z^n$, convergence is guaranteed only when z remains strictly within the largest modulus of these roots, in order to avoid the singularities caused by the vanishing of the determinant.

In this case too, we can attempt to represent the inverse as a power series:

$$A(z)^{-1} = \sum_{n=0}^{\infty} C_n z^n$$

However, this representation is only valid within a certain radius of convergence. As in the scalar case, the convergence depends on the location of the roots of the polynomial $\det(A(z))$. Specifically, the series converges for:

$$|z| < R \quad \text{where} \quad R = \frac{1}{\max_i |z_i|}$$

where the z_i 's are the roots of the characteristic polynomial, i.e., the complex numbers for which $\det(A(z)) = 0$. In other words, the series converges as long as z remains strictly inside the disk defined by the largest characteristic root in modulus.

Conclusion

The radius of convergence $R < 1$ if and only if the $\text{VAR}(k)$ process is stable.

Point iv)

Use the equality $A(z)A(z)^{-1} = I_p$ to find a recursive expression of C_i , $i = 0, 1, \dots$, in terms of A_i and previous C_i coefficients.

Proof

We are given the matrix polynomial and its inverse:

$$A(z) = I_p - \sum_{i=1}^k A_i z^i, \quad A(z)^{-1} = \sum_{j=0}^{\infty} C_j z^j$$

We want to derive a recursive expression for the coefficients C_j using the identity:

$$A(z)A(z)^{-1} = I_p$$

Substituting the expressions, we obtain:

$$\left(I_p - \sum_{i=1}^k A_i z^i \right) \left(\sum_{j=0}^{\infty} C_j z^j \right) = I_p$$

Expanding the product:

$$\begin{aligned} \sum_{j=0}^{\infty} C_j z^j - \left(\sum_{i=1}^k A_i z^i \right) \left(\sum_{j=0}^{\infty} C_j z^j \right) &= I_p \\ \sum_{j=0}^{\infty} C_j z^j - \sum_{i=1}^k \sum_{j=0}^{\infty} A_i C_j z^{i+j} &= I_p z^0 + z^1 0 + z^2 0 + z^3 0 \dots \end{aligned}$$

Now we collect terms of equal powers of z .

Term in z^0 : The constant term on the left side is obtained from $j = 0$ in the first summation:

$$C_0 = I_p$$

Terms in z^n , for $n \geq 1$: From the first sum we have $C_n z^n$, and from the double sum we extract terms such that $i + j = n$, i.e., $j = n - i$. Hence:

$$\sum_{i=1}^{\min(n,k)} A_i C_{n-i}$$

So the total coefficient of z^n on the left-hand side becomes:

$$C_n - \sum_{i=1}^{\min(n,k)} A_i C_{n-i}$$

Since the right-hand side is the identity matrix I_p , and contributes only to the z^0 term, we require that all higher-order coefficients vanish:

$$C_n - \sum_{i=1}^{\min(n,k)} A_i C_{n-i} = 0 \quad \text{for } n \geq 1$$

Final recursive formula:

$$C_0 = I_p$$

$$C_n = \sum_{i=1}^{\min(n,k)} A_i C_{n-i}, \quad n \geq 1$$

This recursive formula allows us to compute C_i for $i = 0, 1, 2, \dots$ in terms of A_i and the previous C_i coefficients.

Conclusion

The coefficients C_n satisfy the recursion:

$$C_0 = I_p, \quad C_n = \sum_{i=1}^{\min(n,k)} A_i C_{n-i}.$$

Point v)

What is the expression of C_i if $k = 1$?

Proof

Let us now analyze the case in which the matrix polynomial has degree $k = 1$, that is:

$$A(z) = I_p - A_1 z \quad \text{with corresponding inverse} \quad A(z)^{-1} = \sum_{n=0}^{\infty} C_n z^n$$

In this setting, the recursive formula derived in the previous point simplifies substantially. Since $k = 1$, the recursive relation becomes:

$$C_n = A_1 C_{n-1}, \quad \text{with initial condition } C_0 = I_p$$

We can now compute the first few coefficients explicitly:

$$\begin{aligned} C_1 &= A_1 C_0 = A_1 \cdot I_p = A_1 \\ C_2 &= A_1 C_1 = A_1 \cdot A_1 = A_1^2 \\ C_3 &= A_1 C_2 = A_1 \cdot A_1^2 = A_1^3 \\ &\vdots \\ C_n &= A_1^n \end{aligned}$$

Therefore, when $k = 1$, the closed-form expression for the coefficients is:

$$C_n = A_1^n \quad \text{for all } n \geq 0$$

Conclusion

When $k = 1$, $C_n = A_1^n$.

Point vi)

With a software of your choice, write a function for the previous recursions.

Results

The implementation in R of the recursive formula for computing the matrices C_n for $n = 0, \dots, N$, given the matrices $A_1, \dots, A_k$ is provided in the Appendix.

Exercise 3

Point i)

State the stability Condition

Proof

For the VAR(1) model $X_t = AX_{t-1} + \varepsilon_t$, the stability condition requires that all eigenvalues of matrix A must lie inside the unit circle (i.e., have modulus less than 1).

Given $A = V\Lambda V^{-1}$, where:

$$V = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \Lambda = \begin{pmatrix} \lambda_1 & 0 \\ 0 & 0.5 \end{pmatrix}$$

The eigenvalues of A are simply the diagonal elements of Λ , which are λ_1 and 0.5. Therefore, the stability condition is:

$$|\lambda_1| < 1 \quad \text{and} \quad |0.5| < 1$$

Conclusions

Since 0.5 is always less than 1, the key condition is $|\lambda_1| < 1$.

Point ii)

Generate Realization with $\lambda_1 = 0.2$

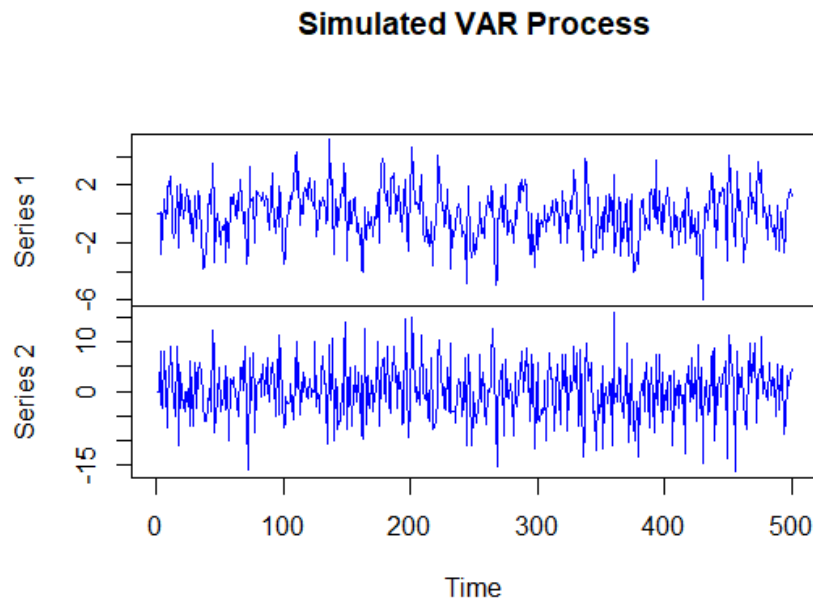


Figure 1: Simulation of VAR(1) with T=500

Point iii)

Estimate A and Ω and Compute Characteristic Roots

Results

\$True_A

```
      [,1] [,2]
[1,]  1.1 -0.3
[2,]  1.8 -0.4
```

\$Estimated_A

```
      [,1] [,2]
[1,] 1.018089 -0.2824145
[2,] 1.645991 -0.3717240
```

\$True_Omega

```
      [,1] [,2]
[1,]    1    3
[2,]    3   25
```

\$Estimated_Omega

```
      y1      y2
y1 1.050034 3.100134
y2 3.100134 23.662814
```

\$True_Eigenvalues

```
[1] 0.2 0.5
```

\$Estimated_Eigenvalues

```
[1] 0.1888565 0.4575084
```

Point iv)

Compute and Plot IRFs with Cholesky Decomposition, using the value of A and Ω in the DGP.

Results

To calculate the IRF, we need to transform the reduced errors (ϵ_t) into structural errors (u_t), such that $\epsilon_t = Pu_t$, where P is the lower triangular matrix obtained from the Cholesky decomposition of Ω :

$$\Omega = DD'$$

The IRF measures the response of the system's variables to a structural shock (u_t). To compute it, we use the VAR coefficients and the matrix P .

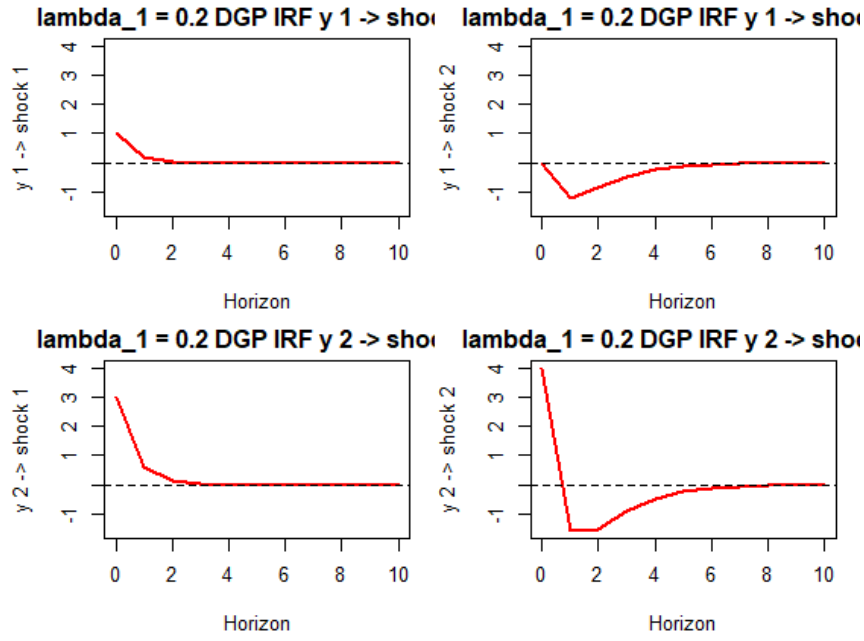


Figure 2: IRF with A and Ω

Point v)

Compute and Plot IRFs with Cholesky Decomposition, using the value of A and Ω in the DGP.

Results

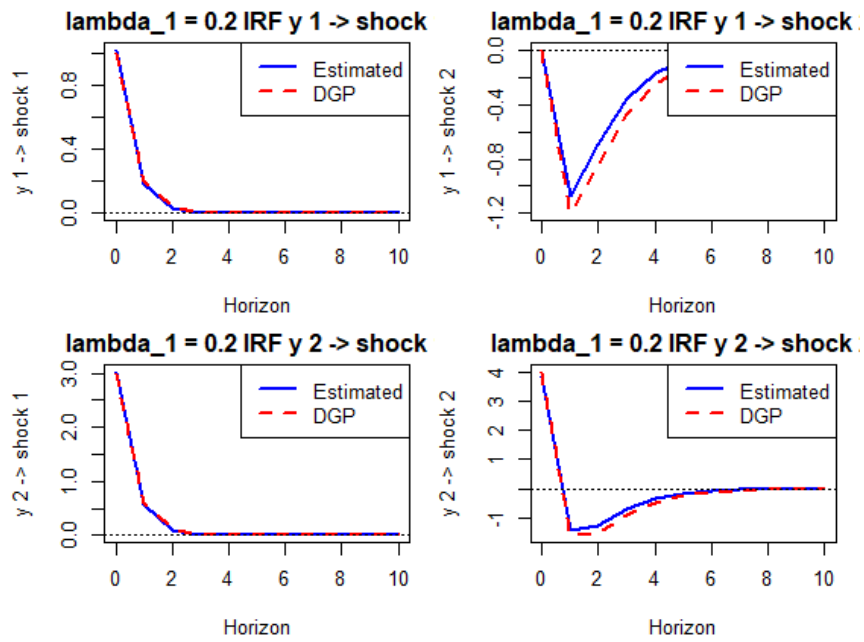


Figure 3: IRF with $\hat{A}$ and $\hat{\Omega}$

Point vi)

Repeat Points ii) - v) getting $\lambda_1 = 0.6$

Results

True A:

```

      [,1] [,2]
[1,]  0.3  0.1
[2,] -0.6  0.8

```

Estimated A:

```

      [,1]      [,2]
[1,] 0.2292900 0.1222175
[2,] -0.6452649 0.8337460

```

True Omega:

```

      [,1] [,2]
[1,]    1    3
[2,]    3   25

```

Estimated Omega:

```

      y1      y2
y1 1.052140 3.107039
y2 3.107039 23.679879

```

True Eigenvalues: 0.6 0.5

Estimated Eigenvalues: 0.6432279 0.4198082

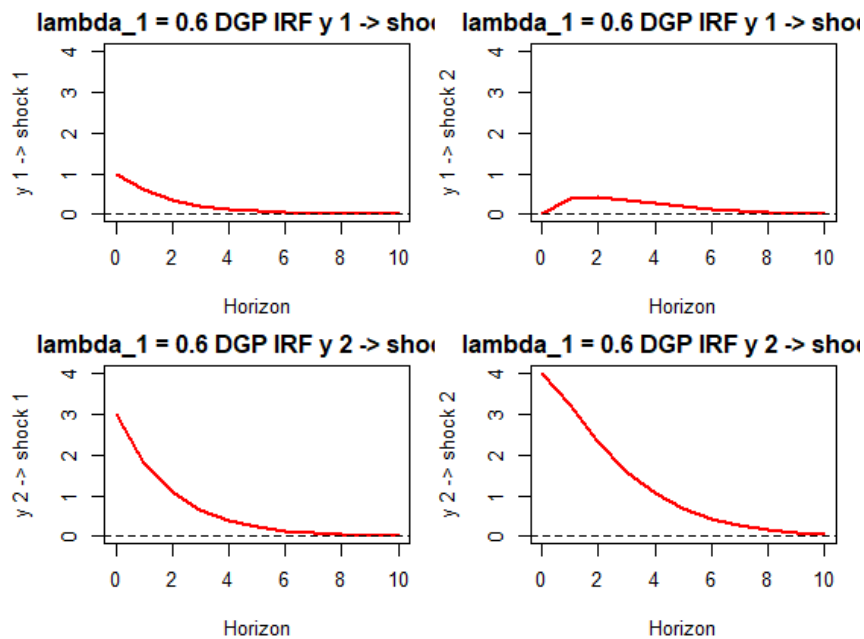


Figure 4: IRF of DGP with $\lambda_1 = 0.6$

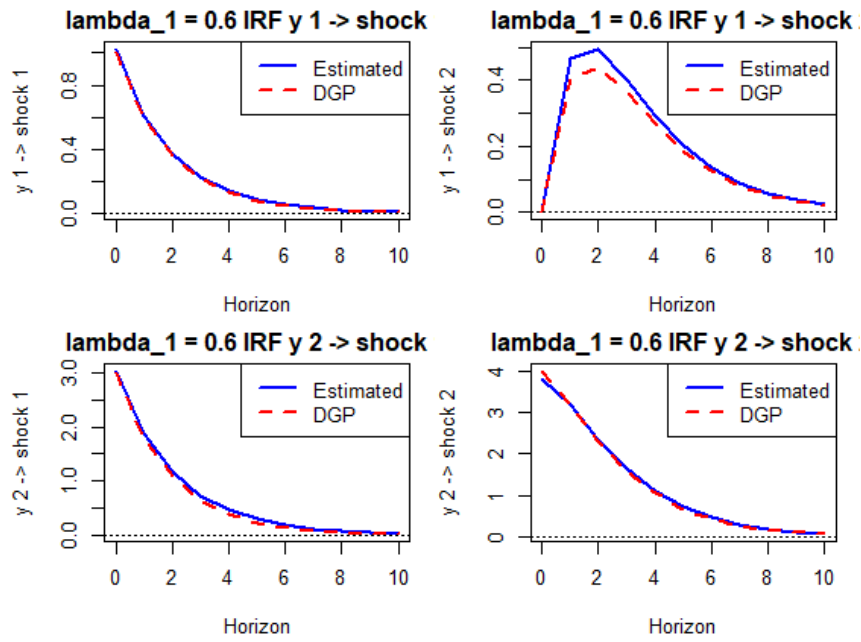


Figure 5: IRF of estimates with $\lambda_1 = 0.6$

Point vii)

Repeat Points ii) - v) getting $\lambda_1 = 1$

Results

True A:

```
[,1] [,2]
[1,] -0.5 0.5
[2,] -3.0 2.0
```

Estimated A:

```
[,1] [,2]
[1,] -0.5633237 0.5214038
[2,] -3.1079662 2.0369395
```

True Omega:

```
[,1] [,2]
[1,] 1 3
[2,] 3 25
```

Estimated Omega:

```
      y1      y2
y1 1.054099 3.117726
y2 3.117726 23.735146
```

True Eigenvalues: 1 0.5

Estimated Eigenvalues: 1.001075 0.4725413

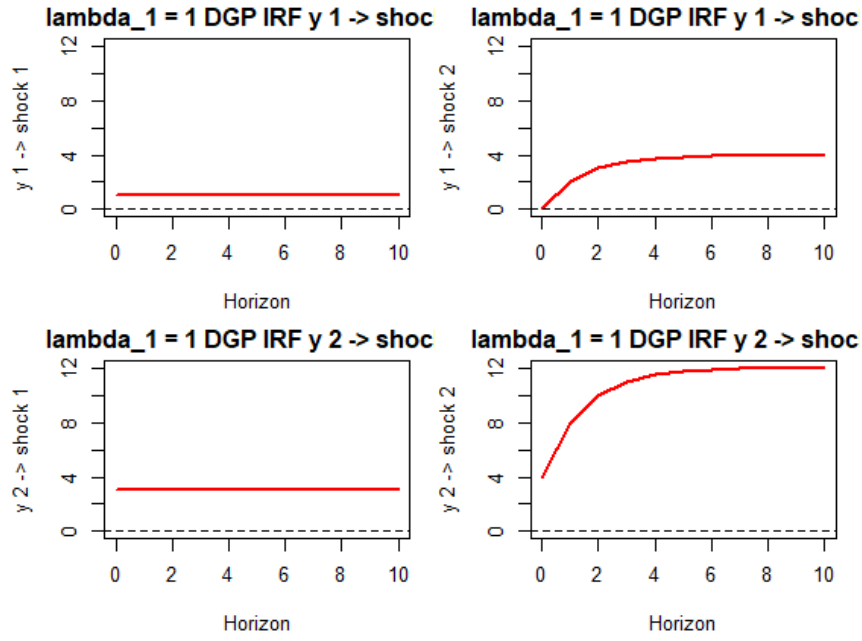


Figure 6: IRF of DGP with $\lambda_1 = 1$

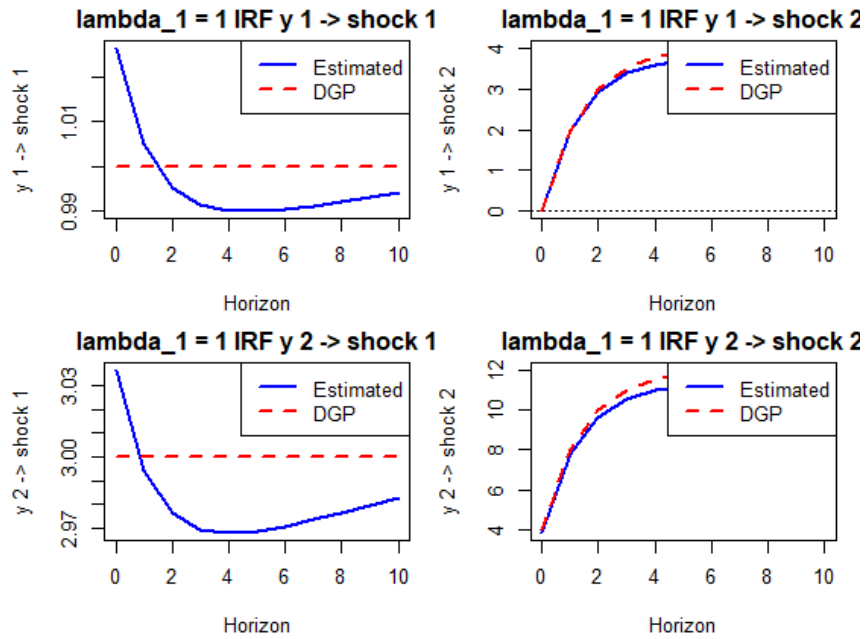


Figure 7: IRF of estimates with $\lambda_1 = 1$

Point viii)

Repeat Points ii) - v) getting $\lambda_1 = 1.4$

Results

True A:

```

      [,1] [,2]
[1,] -1.3  0.9
[2,] -5.4  3.2

```

Estimated A:

```

      [,1] [,2]
[1,] -1.299992 0.8999976
[2,] -1.236814 1.4845072

```

True Omega:

```

      [,1] [,2]
[1,]    1    3
[2,]    3   25

```

Estimated Omega:

```

      y1      y2
y1    1.059968 -147.0742
y2 -147.074217 4585859.7393

```

True Eigenvalues: 1.4 0.5

Estimated Eigenvalues: 1.000679 -0.8161632

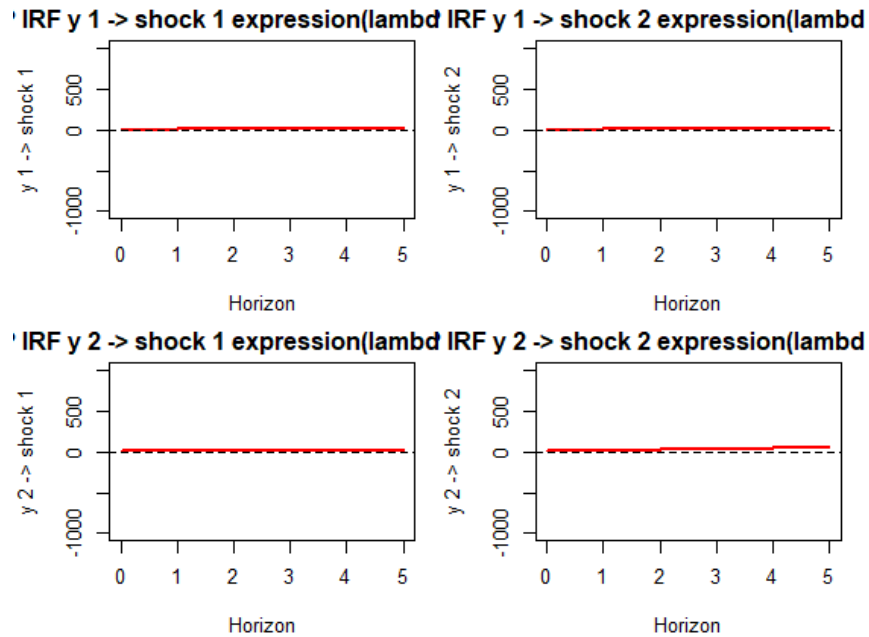


Figure 8: IRF of DGP with $\lambda_1 = 1.4$

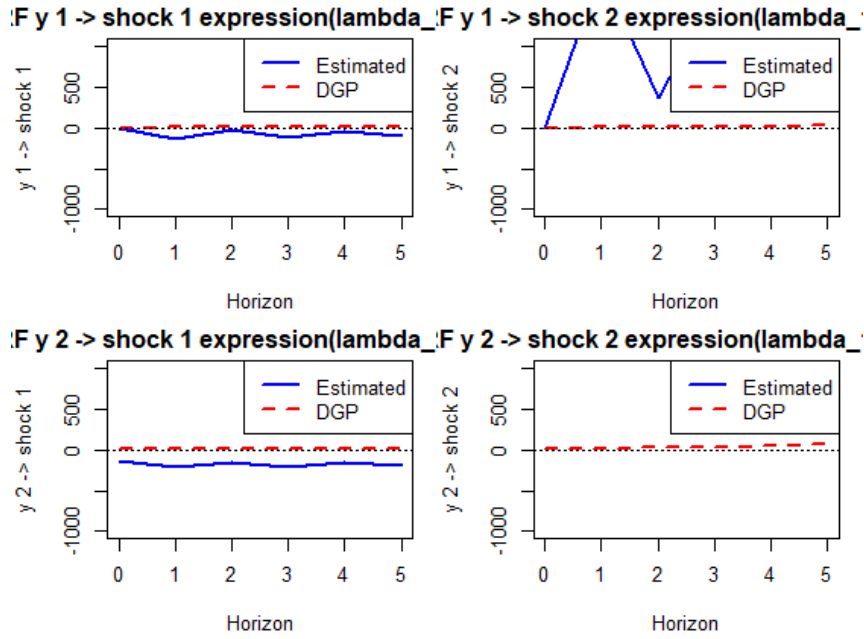


Figure 9: IRF of estimates with $\lambda_1 = 1.4$

Conclusion

The VAR model is unstable when $|\lambda_1| > 1$. Here, with $\lambda_1 = 1.4$, an eigenvalue of A exceeds 1, causing explosive behavior in X_t . This may lead to extremely large values or Inf/NaN in simulations, IRFs, or plots.

Point ix)

Comment on Results

Conclusions

The results vary with respect to λ_1 as follows:

- **When $|\lambda_1| < 1$ (stable cases, e.g., $\lambda_1 = 0.2, 0.6$):**
 - The estimated parameters are close to the DGP values.
 - The impulse response functions (IRFs) decay to zero as the shock dissipates.
 - The closer λ_1 is to 1, the more persistent the IRFs become.
- **When $\lambda_1 = 1$ (unit root case):**
 - The process becomes non-stationary.
 - The IRFs do not decay but remain constant over time.
 - Estimation becomes less precise.

- **When $\lambda_1 = 1.4$ (explosive case):**
 - The process becomes explosive.
 - The IRFs grow without bound.
 - Estimation becomes unreliable as the variance increases over time.

Exercise 4

Point i)

Show D is a Decomposition of Ω iff $D = D_0 Q$

Proof

A decomposition D of Ω satisfies the condition $\Omega = DD'$. The Cholesky decomposition D_0 is a special case where D_0 is lower triangular and satisfies $\Omega = D_0 D_0'$. This choice guarantees a unique decomposition with a lower triangular structure.

For any orthogonal matrix Q ($QQ' = I$), if we define $D = D_0 Q$, then:

$$DD' = D_0 Q Q' D_0' = D_0 I D_0' = D_0 D_0' = \Omega.$$

Conversely, for any valid decomposition D satisfying $DD' = \Omega$, we can express D in terms of D_0 as $D = D_0 Q$, where:

$$Q = D_0^{-1} D.$$

To verify that Q is indeed orthogonal, we check:

$$QQ' = (D_0^{-1} D)(D_0^{-1} D)' = D_0^{-1} D D' (D_0^{-1})' = D_0^{-1} \Omega (D_0^{-1})' = D_0^{-1} D_0 D_0' (D_0^{-1})' = I.$$

Conclusion

Since $QQ' = I$, we conclude that Q is an orthogonal matrix, confirming that any decomposition D must be of the form $D = D_0 Q$.

Point ii)

Show Likelihood is the Same for All D 's

Proof

The likelihood function depends on Ω via:

$$-\frac{T}{2} \ln |\Omega| - \frac{1}{2} \text{tr}(\Omega^{-1} S),$$

where S is the sample covariance matrix. Since we have established that any valid D can be expressed as $D = D_0 Q$ for some orthogonal Q , we examine its effect on the likelihood.

For any D satisfying $\Omega = DD'$, we observe that:

$$|\Omega| = |DD'| = |D_0 Q Q' D_0'| = |D_0 D_0'|,$$

where we use the fact that $|Q| = 1$ since Q is orthogonal. Thus, $|\Omega|$ remains unchanged.

Similarly, for the trace term:

$$\text{tr}(\Omega^{-1} S) = \text{tr}((DD')^{-1} S) = \text{tr}((D_0 Q Q' D_0')^{-1} S) = \text{tr}((D_0 D_0')^{-1} S).$$

Conclusion

Since neither $|\Omega|$ nor $\text{tr}(\Omega^{-1}S)$ depends on the choice of Q , the likelihood remains unchanged across different decompositions D , confirming its invariance.

Point iii)

Relevance for Analysis

Results

Different choices of D lead to different impulse responses, even though they yield the same likelihood. This highlights that the Cholesky decomposition is a convention rather than a necessity. However, it is not the only possible choice. Any decomposition $D = D_0Q$ with an orthogonal matrix Q produces an alternative representation of the same covariance structure, leading to different but observationally equivalent structural interpretations. This highlights that while statistical inference based on the likelihood remains unchanged, economic or structural interpretations can differ depending on the choice of D .

Exercise 5

Point i) - ii)

Consider the DGP in Exercise 3 with $\lambda_1 = 0.6$; with a software of your choice, generate 3 Q 's and compute the corresponding IRFs. Is this is relevant for the analysis?

Results

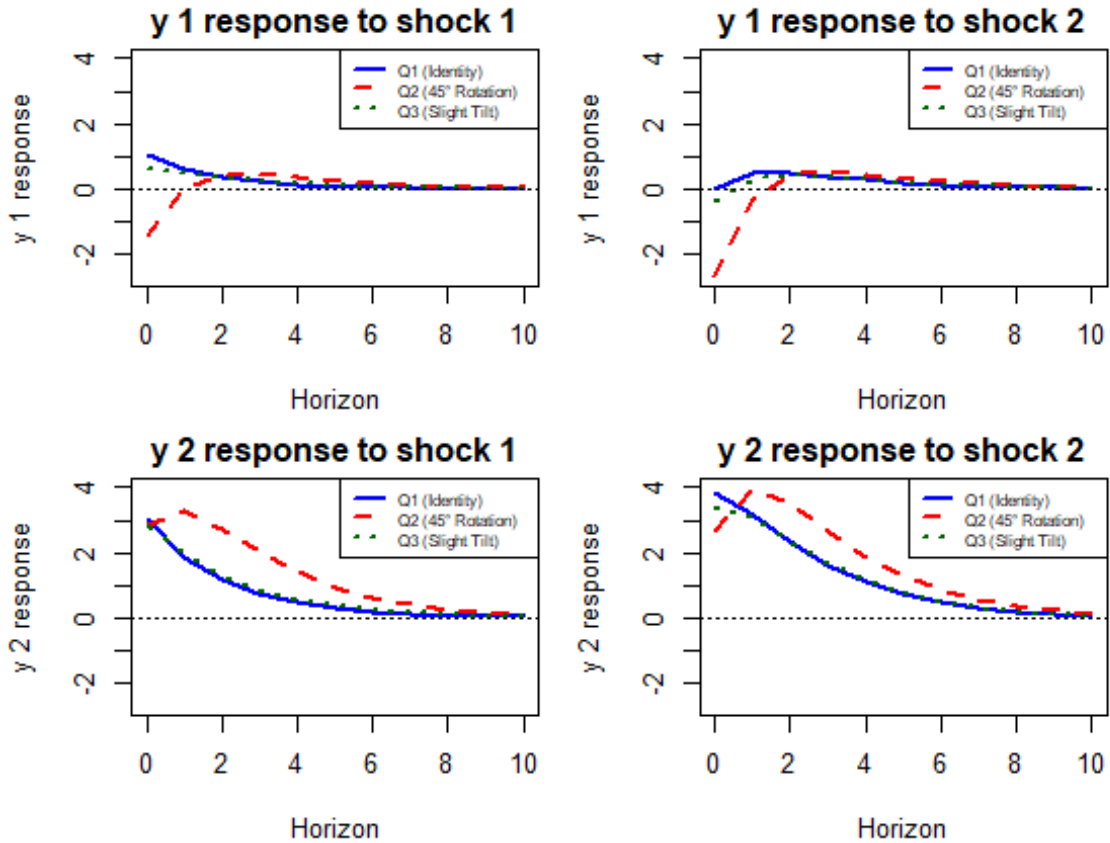


Figure 10: IRFs from a VAR(1) model under three different orthogonal identification schemes: **Q1** (Identity), **Q2** (45° Rotation), and **Q3** (Slight Tilt).

Although the underlying VAR model is the same across all cases—meaning the likelihood function and estimated parameters of the VAR remain unchanged—the choice of the orthogonal rotation matrix Q used in structural identification crucially affects the shape of the IRFs.

- **Q1 (Identity)**: This is the standard Cholesky identification where the shocks are interpreted as orthogonal innovations without rotation. It serves as the benchmark IRF.

- **Q2 (45° Rotation):** A symmetric rotation mixes the original structural shocks. As seen in the plots, this alters the direction and magnitude of the responses dramatically, despite preserving orthogonality.
- **Q3 (Slight Tilt):** A minor perturbation to the identity matrix still leads to visible differences in the IRF paths, especially in the persistence and cross-variable effects.

This highlights a key point in structural VAR analysis: **the IRFs are not uniquely identified from the data alone.** While the reduced-form VAR is statistically identified and its estimation consistent, the transformation to structural IRFs requires an identifying assumption—here expressed through the choice of Q . Thus, different economically meaningful IRFs can be derived from the same estimated VAR by changing the identification scheme.

Conclusion

Structural IRFs depend on identification assumptions.

Exercise 6

Let $X_t = AX_{t-1} + \varepsilon_t$, with characteristic roots z_i , $i = 1, 2, \dots, n$;

Point i)

Sum and subtract X_{t-1} and rewrite the VAR as $\Delta X_t = \Pi X_{t-1} + \varepsilon_t$, where $\Delta = 1 - L$ is the difference operator and $\Pi = A - I$.

Proof

As suggested, let us start summing and subtracting X_{t-1} :

$$\begin{aligned} X_t &= AX_{t-1} - X_{t-1} + X_{t-1} + \varepsilon_t = (A - I)X_{t-1} + \varepsilon_t + X_{t-1} \\ \Rightarrow X_t - X_{t-1} &= (A - I)X_{t-1} + \varepsilon_t \end{aligned}$$

Let's recall the Lag operator $L^k X_t = X_{t-k}$. In particular, with $k=1$, we obtain $LX_t = X_{t-1}$ and this quantity can be used in the previous equation:

$$\begin{aligned} X_t - LX_t &= (A - I)X_{t-1} + \varepsilon_t \\ (1 - L)X_t &= (A - I)X_{t-1} + \varepsilon_t \\ \Delta X_t &= \Pi X_{t-1} + \varepsilon_t \end{aligned}$$

That is exactly the final form requested by the exercise.

Point ii)

Show that $\text{rank } \Pi = p$ if and only if no characteristic root is equal to one.

Proof

Since A is a $p \times p$ matrix, $\Pi = A - I$ is also a $p \times p$ matrix. Therefore, its maximum possible rank is p .

Hence, the purpose of this point is to verify that Π has full rank if and only if no characteristic root z_i is equal to 1.

Firstly, let's demonstrate ($\Rightarrow$):

We know that $\Pi = A - I$ and that $\text{rank}(\Pi) = p$. Therefore, Π has full rank and is invertible (a matrix is invertible if and only if it has full rank, since this implies that its determinant is non-zero).

As a consequence, $A = \Pi + I$ is the sum of an invertible matrix and the identity matrix.

If one of the eigenvalues of A were equal to 1, then Π would have a zero eigenvalue, which would make it non-invertible and thus of rank less than p .

But this contradicts our assumption that $\text{rank}(\Pi) = p$, and therefore the forward implication of the “if and only if” statement is verified.

Now let's demonstrate ($\Leftarrow$):

If none of the characteristic roots is equal to 1, this means that 1 is not an eigenvalue of A , and therefore $\Pi = A - I$ has no zero eigenvalues.

As a result, we can conclude that Π is invertible and has full rank.

Hence, the backward direction of the “if and only if” statement is confirmed.

Conclusion

We have thus proved both directions of the equivalence, so $\text{rank } \Pi = p$ if and only if no characteristic root is equal to one.

Point iii)

Show that $\text{rank } \Pi = 0$ if and only if all characteristic roots are equal to one.

Proof

Firstly, let's prove the forward direction ($\Rightarrow$):

If the rank of a matrix M is zero, this means that all of its entries are equal to zero. Therefore, the following relation holds:

$$\Pi = A - I \Rightarrow A - I = 0 \Rightarrow A = I$$

In other words, A is the identity matrix; and an identity matrix has all its eigenvalues equal to 1.

Thus, if $\text{rank}(\Pi) = 0$ and $A = I$, then all the characteristic roots (i.e., the eigenvalues of A) are equal to 1.

Now let's prove the backward direction ($\Leftarrow$):

Conversely, if all the characteristic roots are equal to 1, then all the eigenvalues of A are 1. As a result, all the eigenvalues of $\Pi = A - I$ are equal to 0. This follows from the fact that the eigenvalues of $\Pi = A - I$ are given by $z_i - 1$, where z_i are the eigenvalues of A .

In conclusion, if all the eigenvalues of a square matrix are zero, then the matrix is not invertible and its rank is 0. Therefore, the backward direction is also proved.

Conclusion

We have proved both directions of the equivalence, so $\text{rank } \Pi = 0$ if and only if all characteristic roots are equal to one.

Point iv)

Show that $\text{rank } \Pi < p$ if and only if some characteristic root is equal to one.

Proof

Let us prove the forward implication ($\Rightarrow$):

If there exists at least one characteristic root such that $z_i = 1$, then $z_i - 1 = 0$, which implies that 0 is an eigenvalue of $\Pi = A - I$.

Therefore (as explained in the previous points), Π is not invertible, and we conclude that $\text{rank}(\Pi) < p$.

Now let us prove the backward implication ($\Leftarrow$):

If $\text{rank}(\Pi) < p$, then Π is not invertible. This means that Π has at least one zero eigenvalue.

As a result, there exists at least one characteristic root of A such that $z_i = 1$, since the eigenvalues of $\Pi = A - I$ are of the form $z_i - 1$.

Conclusion

Both sides of the if and only if relation have been proved, so $\text{rank } \Pi < p$ if and only if some characteristic root is equal to one.

Exercise 7

Work with the file `MSCI_daily.xlsx` contains the daily time series of the MSCI world index and its 11 GICS sectors.

Point i)

With a software of your choice, i) upload data and tranform the series in logs.

Results

The dataset was successfully uploaded and all series were transformed into log-values for analysis. The complete R code used for data processing is provided in the Appendix.

Point ii)

Plot the series and their acf and comment.

Results

The plots below display the 12 log-transformed time series and their corresponding autocorrelation functions (ACF).

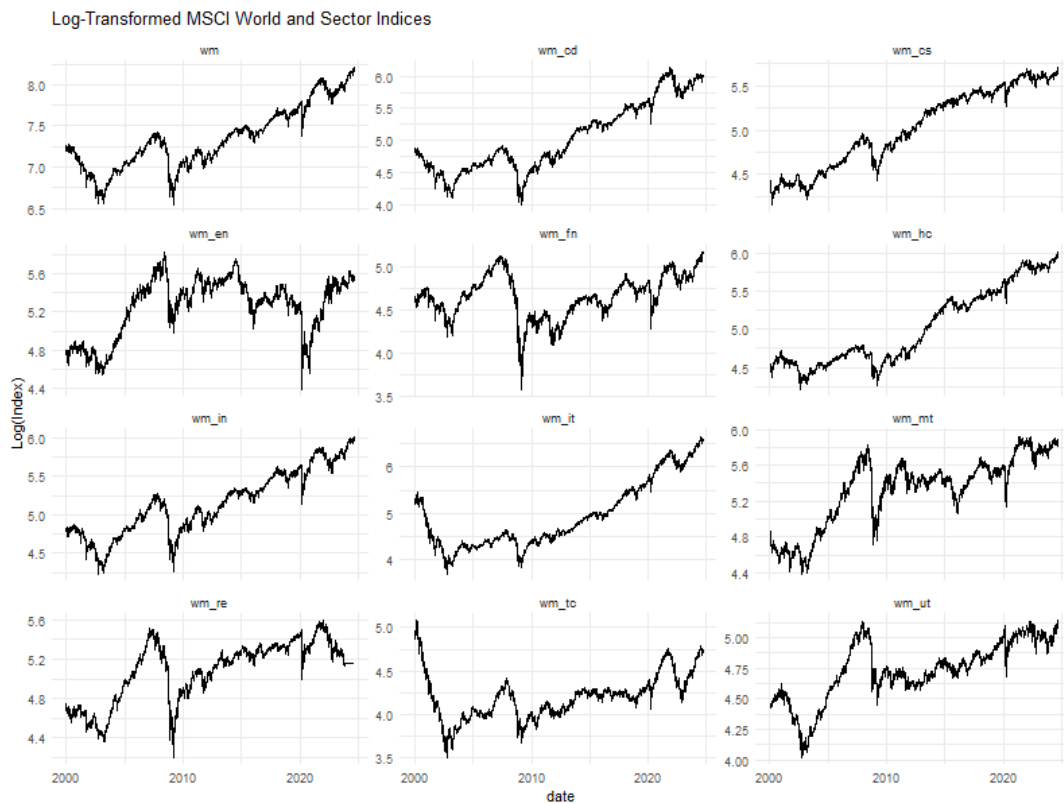


Figure 11: Log-Transformed Time Series of MSCI Indices

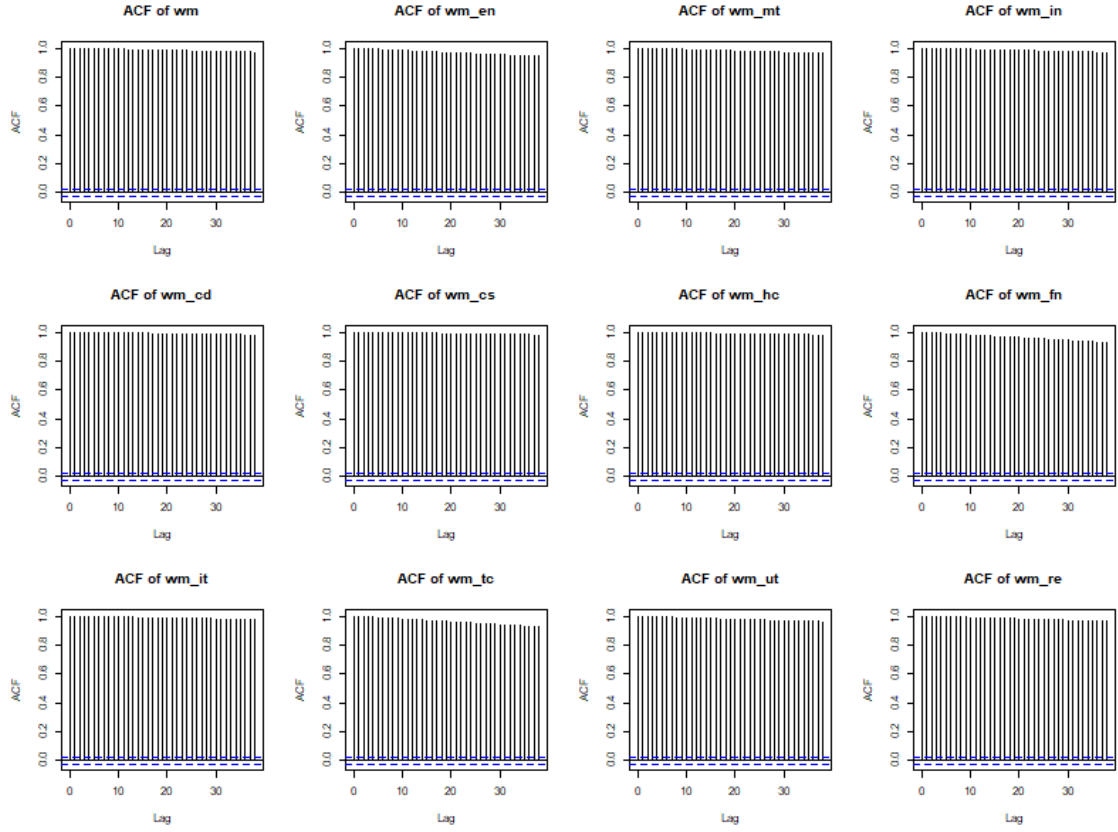


Figure 12: Autocorrelation Functions (ACF) of Log Series

The time series plots in Figure 11 exhibit clear signs of non-stationarity, with persistent trends and visible structural movements. This visual impression is confirmed in Figure 12, where the autocorrelation functions (ACFs) of all series remain close to 1 even at high lags.

Such slow decay in the ACF is a classical signature of non-stationary processes, typically associated with the presence of unit roots. In these cases, shocks to the series have long-lasting or permanent effects, which undermines the assumption of weak dependence necessary for many standard time series methods. This empirical evidence suggests that further transformations or differencing may be required before reliable VAR modeling can be conducted.

Point iii)

Estimate VAR(1) model on 3 indices of your choice and run the residual diagnostics. Is this a reasonable model? If not, add one lag and check if diagnostics improve.

Results

Estimation of a VAR(1) model was performed on the first three indices, namely: `wm`, `wm_en`, and `wm_mt`.

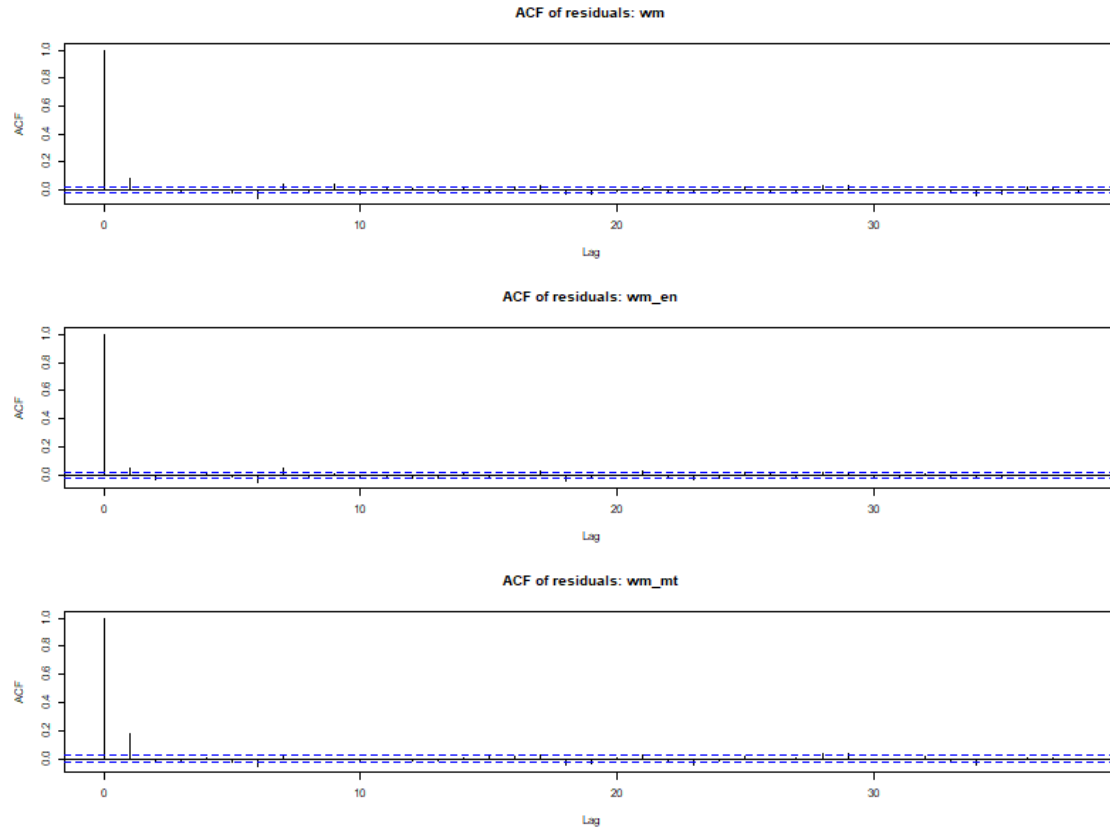


Figure 13: Autocorrelation Functions (ACF) of Residuals

The residual diagnostics for the VAR(1) model highlight several important concerns:

- **Autocorrelation:** While the ACF plots of residuals appear largely uncorrelated individually, the Portmanteau test strongly rejects the null of no joint autocorrelation. This implies significant multivariate residual dependence and possible model misspecification.
- **Non-normality:** The results of the Jarque-Bera test, as well as separate skewness and kurtosis components, clearly reject the assumption of multivariate normality. This undermines the validity of conventional statistical inference.
- **Heteroskedasticity:** The multivariate ARCH test reveals strong evidence of conditional heteroskedasticity in the residuals, indicating the presence of time-varying volatility.

- **Contemporaneous Correlation:** The residual correlation matrix displays high off-diagonal values, indicating strong contemporaneous dependence across equations.

VAR(2) Model Assessment:

To improve model fit, a VAR(2) model was estimated. However, the residual diagnostics still indicate significant specification issues:

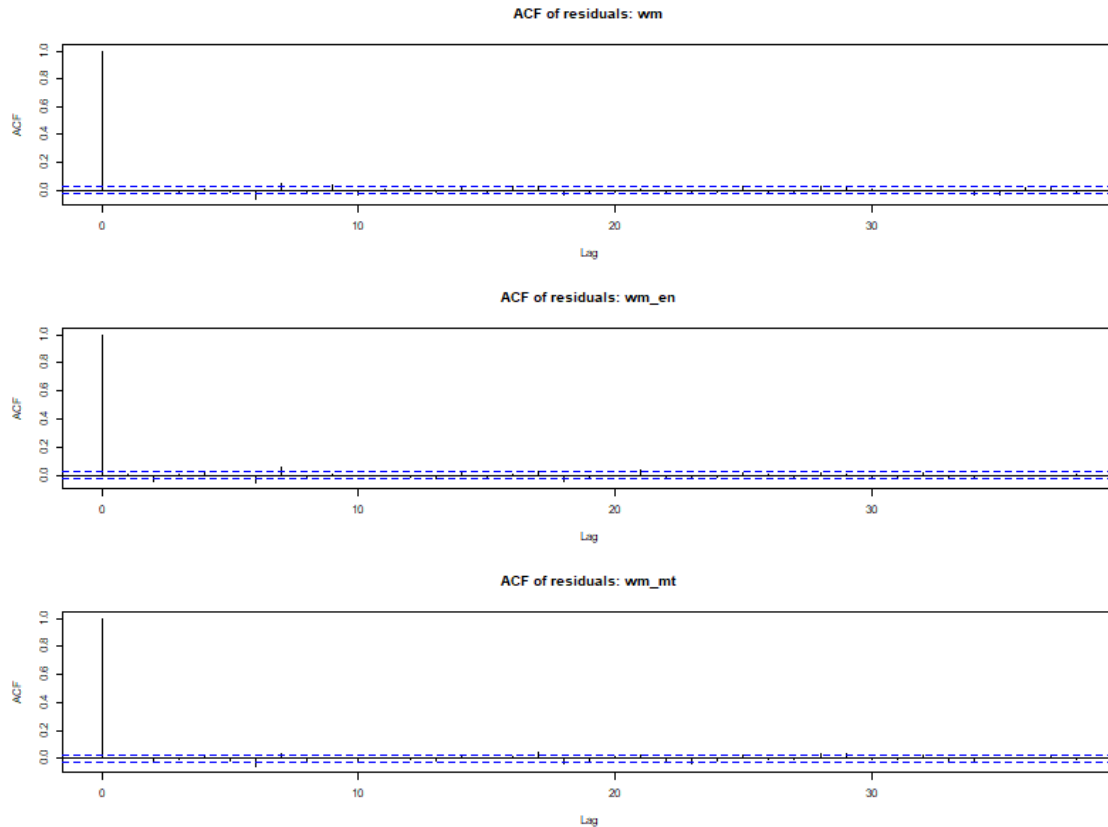


Figure 14: Autocorrelation Functions (ACF) of Residuals

- **Residual Autocorrelation:** The Portmanteau test continues to reject the null hypothesis, implying the presence of multivariate serial correlation even with two lags.
- **Non-normality and Heteroskedasticity:** The Jarque-Bera and ARCH tests again strongly reject the assumptions of normally distributed and homoskedastic residuals.
- **Residual Correlation:** Residuals remain highly contemporaneously correlated, indicating the dynamics between series are still not fully captured.

Conclusion

Increasing the lag order to 2 does not sufficiently resolve the model's shortcomings. The persistent diagnostic failures suggest that the under-

lying issue is likely due to nonstationarity in the data rather than an insufficient lag structure.

Point iv)

Compute the characteristic roots and discuss stability.

Results

To assess the dynamic stability of the VAR models, we examine the eigenvalues of the companion matrix. Stability requires that all such roots lie strictly inside the unit circle in the complex plane.

- **VAR(1) Eigenvalues:** 0.999976, 0.999197, 0.997634
- **VAR(2) Eigenvalues:** 0.999901, 0.998490, 0.998490, 0.112901, 0.056535, 0.056535

All roots are technically within the unit circle; however, several are extremely close to the boundary. In particular:

- The VAR(1) roots are dangerously close to 1, suggesting the presence of unit roots or near-unit-root behavior.
- The VAR(2) model includes some smaller roots, its largest eigenvalues remain nearly unitary, indicating persistent nonstationary dynamics.

Conclusion

From a stability perspective, neither VAR(1) nor VAR(2) models in levels can be considered dynamically stable in a strict sense. This further supports the presence of nonstationarity..

Point v)

Repeat iii) with other 3 indices.

Results

The VAR analysis was repeated using a different set of indices: `wm_tc`, `wm_ut`, and `wm_re`.

- The VAR(1) and VAR(2) models showed high R^2 and strongly significant own-lag coefficients, similar to the earlier group.
- However, residual diagnostics again revealed violations of key assumptions:
 - Strong residual autocorrelation (Portmanteau test rejection)
 - Non-normality (Jarque-Bera, skewness, and kurtosis all rejected)
 - Heteroskedasticity (ARCH test rejected)

- High contemporaneous correlation among residuals
- **VAR(1) Eigenvalues:** 0.999096, 0.999096, 0.997007
- **VAR(2) Eigenvalues:** 0.999027, 0.999027, 0.996868, 0.102345, 0.102345, 0.022702
- The eigenvalues of both VAR(1) and VAR(2) companion matrices are extremely close to 1 , implying the persistence of nonstationary dynamics.

Conclusion

Similar to the first group of indices, this second VAR application confirms that estimation in levels results in unstable models with residuals violating core assumptions.

Point vi)

Repeat iii) with all 11 sectors.

Results

A comprehensive VAR model was estimated using the full set of sector indices. The results are consistent with prior findings:

- **High In-Sample Fit:** All equations exhibit very high R^2 values , and most own-lag coefficients are statistically significant and near one.
- **Residual Issues Persist:**
 - The Portmanteau test continues to strongly reject the null of no residual autocorrelation.
 - The multivariate Jarque-Bera test indicates substantial departures from normality in both skewness and kurtosis.
 - The ARCH-LM test provides strong evidence of heteroskedasticity.
 - Residuals exhibit high contemporaneous correlations, suggesting strong cross-equation dependencies.
- **Stability Concerns:** The eigenvalues of the companion matrices for both VAR(1) and VAR(2) models are extremely close to the unit circle, indicating the presence of unit roots or near-nonstationary behavior.

Conclusion

The full-system VAR analysis reinforces earlier conclusions: estimation in levels leads to model instability and residuals that violate fundamental assumptions of the VAR framework.

Exercise 8

Repeat ii) - vii) in Exercise 7 considering the log-returns of the selected indices.

Point i)

The previous time series have been transformed in log-returns. The following plots below display the log-returns time series and their corresponding autocorrelation functions (ACF)

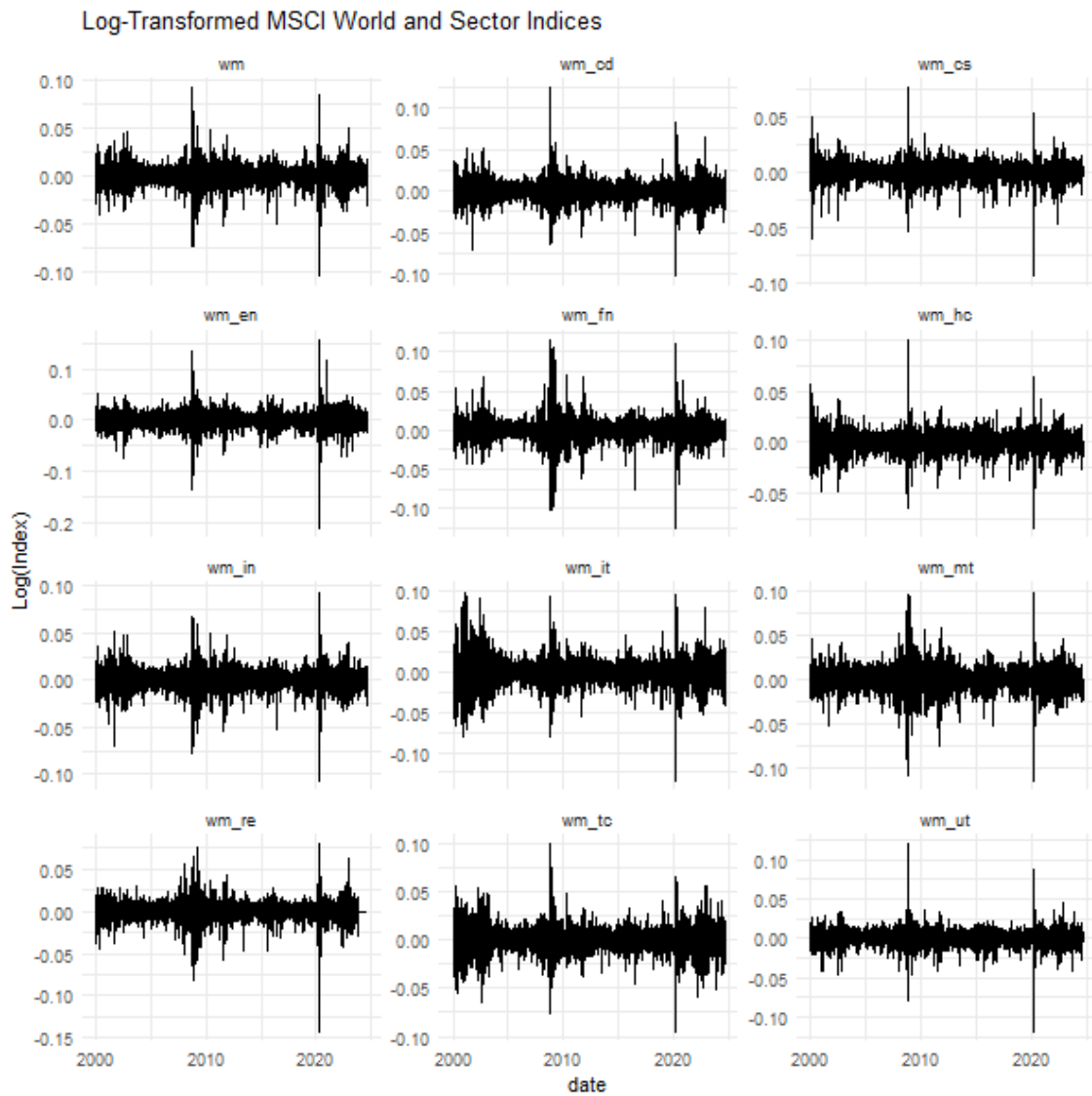


Figure 15: Log-Returns

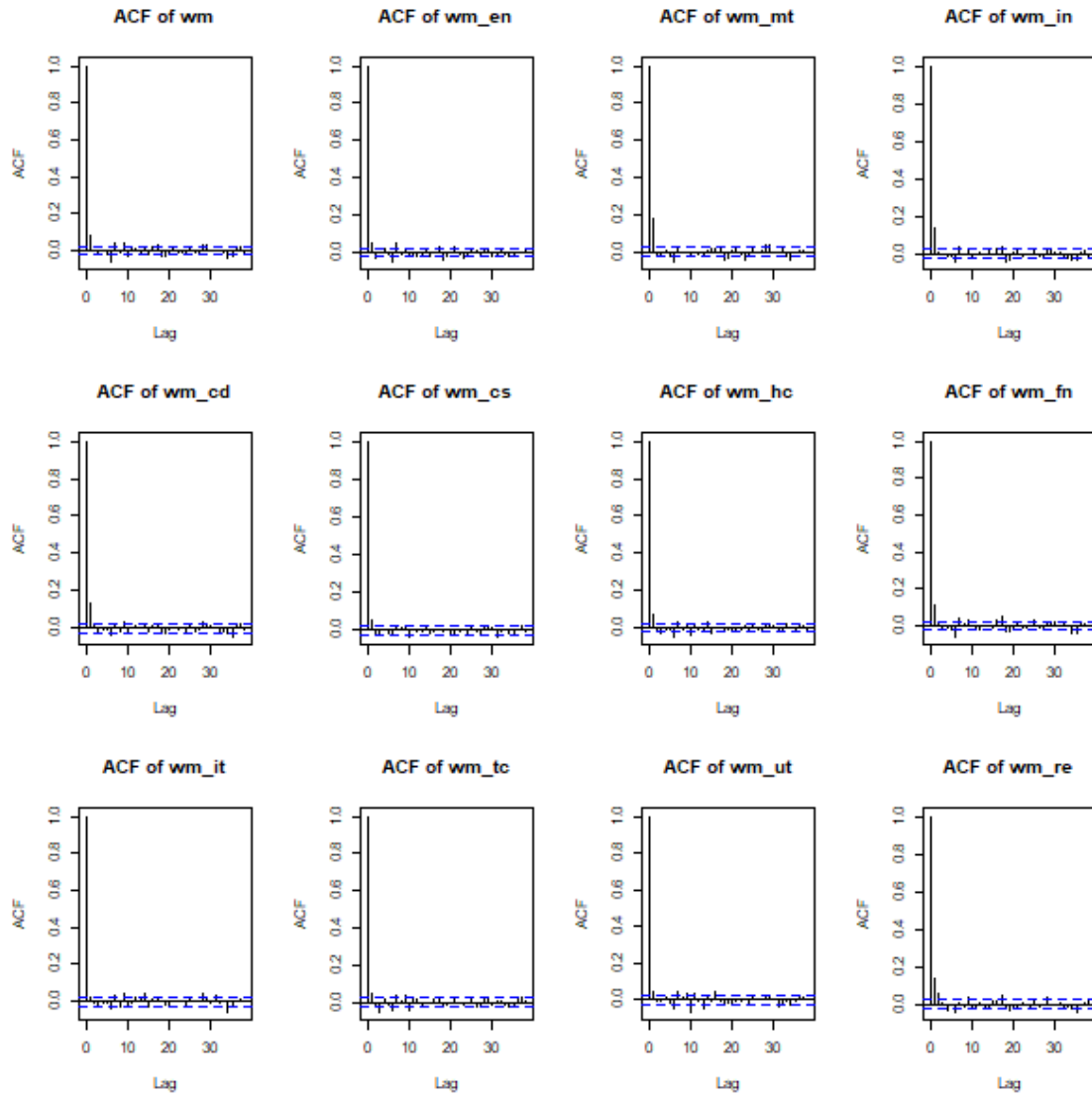


Figure 16: Autocorrelation Functions (ACF) of Log-returns

The time series plots in Figure 15 display the log returns of the time series. Visually, all series appear stationary, with fluctuations oscillating around a constant mean and no visible deterministic trend.

This impression is supported by the autocorrelation functions shown in Figure 16. For most series, only the firsts two lags (and occasionally the third) is statistically significant.

Point ii)

The VAR analysis was conducted on the log return series of the indices: `wm_en`, `wm_mt`, and `wm_in`. Models of lag order 1 and 2 were estimated and compared.

- The estimated coefficients across both VAR(1) and VAR(2) models confirm the relative importance of own-lag dynamics which coefficients remain statistically

significant in both specifications. Cross-lag effects are generally weak and mostly insignificant.

- In terms of dynamic stability, the eigenvalues of the companion matrices for both models are well inside the unit circle:
 - **VAR(1) Eigenvalues:** 0.1292, 0.0645, 0.0645
 - **VAR(2) Eigenvalues:** 0.2835, 0.2835, 0.1759, 0.1759, 0.1313, 0.1313

indicating a clearly stationary process, confirming that differencing the original index levels has resolved the issue of unit roots.

- However, despite this improvement in stationarity, residual diagnostics continue to highlight key misspecification issues:
 - **Residual autocorrelation:** Both models are rejected by the Portmanteau test, indicating persistent multivariate autocorrelation not fully captured by either lag order.
 - **Non-normality:** The multivariate Jarque-Bera tests strongly reject the assumption of residual normality. Skewness and kurtosis components show very large test statistics, indicating both asymmetry and fat tails.
 - **Heteroskedasticity:** The ARCH-LM tests show significant conditional heteroskedasticity in both cases, with extremely small p -values.
 - **Contemporaneous correlation:** The residual correlation matrices reveal strong dependencies across equations, with off-diagonal values consistently above 0.70.

Conclusion

Although log differencing ensures stationarity by stabilizing eigenvalues, the residual issues persist. Both lag orders fail to fully address serial correlation, non-normality, and heteroskedasticity.

Point iii)

The VAR analysis was repeated using a different set of indices: `wm_tc`, `wm_ut`, and `wm_re`.

- The VAR(1) and VAR(2) models confirmed the presence of significant own-lag effects.
- However, the residual diagnostics once again highlighted key violations of VAR assumptions:
 - Strong residual autocorrelation (Portmanteau test rejected the null of no serial correlation in both cases)
 - Non-normality of residuals was strongly rejected by the multivariate Jarque-Bera test, driven by both excess kurtosis and non-zero skewness

- Evidence of conditional heteroskedasticity (ARCH effects) was highly significant
- Residuals displayed strong contemporaneous correlation, particularly between `wm_ut` and `wm_re`
- From a dynamic stability perspective, the eigenvalues of the companion matrices are comfortably within the unit circle:
 - **VAR(1) Eigenvalues:** 0.1036, 0.1036, 0.0238
 - **VAR(2) Eigenvalues:** 0.2809, 0.216, 0.216, 0.1826, 0.150, 0.150

These values confirm that the return series are stationary and that differencing the original level data was appropriate.

Conclusion

Although the differenced series provide stationarity and stable dynamics, the estimated VAR models still suffer from serious residual misspecifications.

Point iv)

A comprehensive VAR model was estimated using the full set of sector indices. The results are consistent with prior findings:

- The Portmanteau test continues to strongly reject the null of no residual autocorrelation, even at lag order 2.
- The multivariate Jarque-Bera test reveals significant non-normality in residuals, driven by both skewness and excess kurtosis.
- The ARCH-LM test identifies strong evidence of residual heteroskedasticity, further confirming that conditional volatility is time-varying.
- The residual correlation matrix displays high contemporaneous correlations, suggesting substantial cross-equation interactions and common shocks across sectors.
- **Stability:** All eigenvalues of the companion matrix lie well inside the unit circle, indicating that the VARs models satisfy the stability condition and do not exhibit signs of explosive or unit root behavior.

Appendix

Exercise 2: R Implementation

```
# Input:
#   A_list: list of matrices A_1, ..., A_k (each of size p x p)
#   N: highest index n for which to compute C_n
# Output:
#   A list of matrices: C_0, C_1, ..., C_N

compute_C <- function(A_list, N) {
  k <- length(A_list)          # Degree of the polynomial (number of A_j terms)
  p <- nrow(A_list[[1]])       # Dimension of the matrices (assumed to be square)
  C_list <- list()             # Will store the matrices C_0 to C_N

  # Initial condition: C_0 = identity matrix I_p
  C_list[[1]] <- diag(p)

  # Recursive computation of C_n for n >= 1
  for (n in 1:N) {
    C_n <- matrix(0, nrow = p, ncol = p) # Start with a zero matrix
    for (j in 1:min(n, k)) {
      # Apply the recurrence: C_n = sum_{j=1}^{min(n,k)} A_j * C_{n-j}
      # (formula found in point iv)
      C_n <- C_n + A_list[[j]] %*% C_list[[n - j + 1]]
    }
    C_list[[n + 1]] <- C_n # Store result
  }

  return(C_list) # Return the full list of matrices
}
```

Exercise 3: R Implementation

```
library(MASS) # For multivariate normal simulation
library(vars) # For VAR estimation

# --- ii) Fix lambda_1 = 0.2 and generate X_t with T = 500 ---
# Define matrices
V <- matrix(c(1, 2, 3, 4), nrow = 2, byrow = TRUE)
Lambda <- matrix(c(0.2, 0, 0, 0.5), nrow = 2, byrow = TRUE)
Omega <- matrix(c(1, 3, 3, 25), nrow = 2, byrow = TRUE)
A <- V %*% Lambda %*% solve(V) # Compute A = V Lambda V^(-1)

# Simulation parameters
T <- 500
```

```

set.seed(123) # For reproducibility
X <- matrix(0, nrow = T, ncol = 2)

# Generate the series
for (t in 2:T) {
  epsilon <- mvrnorm(1, mu = c(0, 0), Sigma = Omega)
  X[t, ] <- A %*% X[t-1, ] + epsilon
}

# Convert to time series and plot
X_ts <- ts(X)
cat("\nnii) Simulated VAR process with T = 500 and lambda_1 = 0.2:\n")
plot(X_ts, main = "Simulated VAR(1) Process", col = c("blue", "red"),
     xlab = "Time", ylab = "Value", lty = 1:2,
     legend = c("y1", "y2"))

# --- iii) Estimate A and Omega, compute characteristic roots ---
var_fit <- VAR(X, p = 1, type = "none")

# Extract A_est correctly
A_est <- matrix(NA, nrow = 2, ncol = 2)
A_est[1,1] <- coef(var_fit)[[1]]["y1.l1", 1] # y1(t-1) in y1 equation
A_est[1,2] <- coef(var_fit)[[1]]["y2.l1", 1] # y2(t-1) in y1 equation
A_est[2,1] <- coef(var_fit)[[2]]["y1.l1", 1] # y1(t-1) in y2 equation
A_est[2,2] <- coef(var_fit)[[2]]["y2.l1", 1] # y2(t-1) in y2 equation

# Extract Omega_est
Omega_est <- summary(var_fit)$covres

# Compute eigenvalues
eigen_A <- eigen(A_est)$values
true_eigen <- c(0.2, 0.5) # True eigenvalues from Lambda

# Print comparison
cat("\nniii) Estimation and comparison with DGP:\n")
cat("True A:\n")
print(A)
cat("Estimated A:\n")
print(A_est)
cat("True Omega:\n")
print(Omega)
cat("Estimated Omega:\n")
print(Omega_est)
cat("True Eigenvalues:", true_eigen, "\n")
cat("Estimated Eigenvalues:", eigen_A, "\n")
cat("Comment: Estimated A and Omega are close to true values due to large T = 500.\n")

```

```

cat(" Eigenvalues are also close to 0.2 and 0.5, confirming good estimation.\n")

# --- iv) Compute and plot IRFs using DGP values ---
D <- t(chol(Omega)) # Cholesky decomposition of Omega

# IRF function
calcola_irf <- function(A, D, horizon) {
  k <- nrow(A)
  irf <- array(0, dim = c(k, k, horizon + 1))
  irf[, , 1] <- D # h=0 is the Cholesky matrix D
  for (h in 1:horizon) {
    irf[, , h + 1] <- A %%% irf[, , h]
  }
  return(irf)
}

# Compute IRF for DGP
horizon <- 5
irf_dgp <- calcola_irf(A, D, horizon)

# Print IRF for DGP
cat("\niv) IRF using DGP values:\n")
for (h in 0:horizon) {
  cat("Horizon", h, ":\n")
  print(irf_dgp[, , h + 1])
}

# Plot IRF for DGP
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for (i in 1:2) {
  for (j in 1:2) {
    plot(0:horizon, irf_dgp[i, j, ], type = "l", col = "red", lwd = 2,
         ylim = range(irf_dgp), xlab = "Horizon",
         ylab = paste("y", i, "-> shock", j),
         main = paste("DGP IRF y", i, "-> shock", j))
    abline(h = 0, lty = 2, col = "black")
  }
}

# --- v) Compute and plot IRFs using estimated values, compare with DGP ---
D_est <- t(chol(Omega_est)) # Cholesky decomposition of Omega_est
irf_est <- calcola_irf(A_est, D_est, horizon)

# Print IRF for estimated values
cat("\nv) IRF using estimated values:\n")
for (h in 0:horizon) {

```

```

    cat("Horizon", h, ":\n")
    print(irf_est[,h + 1])
}

# Plot comparison
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for (i in 1:2) {
  for (j in 1:2) {
    plot(0:horizon, irf_est[i,j,], type = "l", col = "blue", lwd = 2,
         ylim = range(c(irf_est[i,j,], irf_dgp[i,j,])),
         xlab = "Horizon",
         ylab = paste("y", i, "-> shock", j),
         main = paste("IRF y", i, "-> shock", j))
    lines(0:horizon, irf_dgp[i,j,], col = "red", lty = 2, lwd = 2)
    abline(h = 0, lty = 3, col = "black")
    legend("topright", legend = c("Estimated", "DGP"), col = c("blue", "red"),
          lty = c(1, 2), lwd = 2)
  }
}

# --- vi) vii) viii) Repeat ii) - v) getting  $\lambda_1 = 0.6, 1, 1.4$ 

# Function to simulate and analyze VAR for a given lambda_1
simulate_and_analyze <- function(lambda_1, case_name) {
  # Define Lambda
  Lambda <- matrix(c(lambda_1, 0, 0, 0.5), nrow = 2, byrow = TRUE)
  A <- V %*% Lambda %*% solve(V)

  # ii) Simulate X_t with T = 500
  T <- 500
  set.seed(123)
  X <- matrix(0, nrow = T, ncol = 2)
  for (t in 2:T) {
    epsilon <- mvrnorm(1, mu = c(0, 0), Sigma = Omega)
    X[t, ] <- A %*% X[t-1, ] + epsilon
  }
  X_ts <- ts(X)
  cat("\n", case_name, " - ii) Simulated VAR process:\n")
  plot(X_ts, main = paste(case_name, "Simulated VAR(1)"), col = c("blue", "red"),
       xlab = "Time", ylab = "Value", lty = 1:2)

  # iii) Estimate A and Omega, compute eigenvalues
  suppressWarnings({
    var_fit <- VAR(X, p = 1, type = "none")
  })
  A_est <- matrix(NA, nrow = 2, ncol = 2)

```

```

A_est[1,1] <- coef(var_fit)[[1]]["y1.11", 1]
A_est[1,2] <- coef(var_fit)[[1]]["y2.11", 1]
A_est[2,1] <- coef(var_fit)[[2]]["y1.11", 1]
A_est[2,2] <- coef(var_fit)[[2]]["y2.11", 1]
Omega_est <- summary(var_fit)$covres
eigen_A <- eigen(A_est)$values
true_eigen <- c(lambda_1, 0.5)
cat("\n", case_name, " - iii) Estimation and comparison:\n")
cat("True A:\n")
print(A)
cat("Estimated A:\n")
print(A_est)
cat("True Omega:\n")
print(Omega)
cat("Estimated Omega:\n")
print(Omega_est)
cat("True Eigenvalues:", true_eigen, "\n")
cat("Estimated Eigenvalues:", eigen_A, "\n")

# iv) IRF with DGP values
D <- t(chol(Omega))
horizon <- 10
irf_dgp <- calcola_irf(A, D, horizon)
cat("\n", case_name, " - iv) IRF using DGP values:\n")
for (h in 0:horizon) {
  cat("Horizon", h, ":\n")
  print(irf_dgp[,h + 1])
}
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for (i in 1:2) {
  for (j in 1:2) {
    plot(0:horizon, irf_dgp[i,j,], type = "l", col = "red", lwd = 2,
         ylim = range(irf_dgp), xlab = "Horizon", ylab = paste("y", i, "-> shock")
         main = paste(case_name, "DGP IRF y", i, "-> shock", j))
    abline(h = 0, lty = 2, col = "black")
  }
}

# v) IRF with estimated values
D_est <- t(chol(Omega_est))
irf_est <- calcola_irf(A_est, D_est, horizon)
cat("\n", case_name, " - v) IRF using estimated values:\n")
for (h in 0:horizon) {
  cat("Horizon", h, ":\n")
  print(irf_est[,h + 1])
}

```

```

par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))
for (i in 1:2) {
  for (j in 1:2) {
    plot(0:horizon, irf_est[i,j,], type = "l", col = "blue", lwd = 2,
         ylim = range(c(irf_est[i,j,], irf_dgp[i,j,])),
         xlab = "Horizon", ylab = paste("y", i, "-> shock", j),
         main = paste(case_name, "IRF y", i, "-> shock", j))
    lines(0:horizon, irf_dgp[i,j,], col = "red", lty = 2, lwd = 2)
    abline(h = 0, lty = 3, col = "black")
    legend("topright", legend = c("Estimated", "DGP"), col = c("blue", "red"),
          lty = c(1, 2), lwd = 2)
  }
}
}

```

```

# --- vi) Repeat ii) - v) setting lambda_1 = 0.6 ---
cat("\nvi) Case with lambda_1 = 0.6:\n")
simulate_and_analyze(0.6, "lambda_1 = 0.6")

```

```

# --- vii) Repeat ii) - v) setting lambda_1 = 1 and lambda_1 = 1.4 ---
cat("\nvii) Case with lambda_1 = 1:\n")
simulate_and_analyze(1, "lambda_1 = 1")

```

```

cat("\nvii) Case with lambda_1 = 1.4:\n")
simulate_and_analyze(1.4, "lambda_1 = 1.4")

```

Exercise 5: R Implementation

```

# Function to compute IRFs with a given rotation matrix Q
compute_irf_with_Q <- function(A, D, Q, horizon) {
  k <- nrow(A)
  irf <- array(0, dim = c(k, k, horizon + 1))
  irf[, , 1] <- Q %*% D # Apply rotation
  for (h in 1:horizon) {
    irf[, , h + 1] <- A %*% irf[, , h]
  }
  return(irf)
}

# Function to simulate VAR(1) and compute IRFs for different Q matrices
simulate_VAR_and_IRFs <- function(lambda_1, case_name) {
  # Define matrices
  V <- matrix(c(1, 2, 3, 4), nrow = 2, byrow = TRUE)
  Lambda <- matrix(c(lambda_1, 0, 0, 0.5), nrow = 2, byrow = TRUE)
  Omega <- matrix(c(1, 3, 3, 25), nrow = 2, byrow = TRUE)

```

```

A <- V %*% Lambda %*% solve(V)

# Simulate data
T <- 500
set.seed(123)
X <- matrix(0, nrow = T, ncol = 2)
for (t in 2:T) {
  epsilon <- mvrnorm(1, mu = c(0, 0), Sigma = Omega)
  X[t, ] <- A %*% X[t - 1, ] + epsilon
}

# Estimate VAR(1) model
var_fit <- VAR(X, p = 1, type = "none")
A_est <- matrix(NA, nrow = 2, ncol = 2)
A_est[1,1] <- coef(var_fit)[[1]]["y1.l1", 1]
A_est[1,2] <- coef(var_fit)[[1]]["y2.l1", 1]
A_est[2,1] <- coef(var_fit)[[2]]["y1.l1", 1]
A_est[2,2] <- coef(var_fit)[[2]]["y2.l1", 1]
Omega_est <- summary(var_fit)$covres

# Define Q matrices
Q1 <- diag(2)
theta <- pi / 4
Q2 <- matrix(c(cos(theta), -sin(theta), sin(theta), cos(theta)), nrow = 2, byrow = TRUE)
Q3 <- matrix(c(0.9, -0.1, 0.1, 0.9),
             nrow = 2, byrow = TRUE)

# Compute IRFs for each Q
D_est <- t(chol(Omega_est))
horizon <- 10
irf_Q1 <- compute_irf_with_Q(A_est, D_est, Q1, horizon)
irf_Q2 <- compute_irf_with_Q(A_est, D_est, Q2, horizon)
irf_Q3 <- compute_irf_with_Q(A_est, D_est, Q3, horizon)

# Plot IRFs: 2x2 layout, one panel per IRF, each panel shows all Qs
par(mfrow = c(2, 2), mar = c(4, 4, 2, 1))

for (i in 1:2) {
  for (j in 1:2) {
    plot(0:horizon, irf_Q1[i, j, ], type = "l", col = "blue", lwd = 2,
         ylim = range(c(irf_Q1, irf_Q2, irf_Q3)),
         xlab = "Horizon", ylab = paste("y", i, "response"),
         main = paste("y", i, "response to shock", j))
    lines(0:horizon, irf_Q2[i, j, ], col = "red", lty = 2, lwd = 2)
    lines(0:horizon, irf_Q3[i, j, ], col = "green", lty = 3, lwd = 2)
    abline(h = 0, lty = 3, col = "black")
  }
}

```

```

        legend("topright",
              legend = c("Q1 (Identity)", "Q2 (45° Rotation)", "Q3 (Slight Tilt)"),
              col = c("blue", "red", "green"), lty = c(1, 2, 3),
              lwd = 2, cex = 0.7)
      }
    }

}

# Run simulations for lambda_1 = 0.6
simulate_VAR_and_IRFs(0.6, "Lambda_1 = 0.6")

```

Exercise 7 and 8: R Implementation

```

#Exercise 7
#Point i): Upload, transform and manage data
data <- read_excel("MSCI_world_daily.xlsx")
ldata <- cbind(data[,1], log(data[, -1]))
log_data_long <- pivot_longer(as.data.frame(ldata),
                             cols = -date,
                             names_to = "Series",
                             values_to = "LogValue")
png("Timeseries.png", width = 800, height = 600)

ggplot(log_data_long, aes(x = date, y = LogValue)) +
  geom_line() +
  facet_wrap(~ Series, scales = "free_y", ncol = 3) +
  labs(title = "Log-Transformed MSCI World and Sector Indices",
       y = "Log(Index)") +
  theme_minimal()
dev.off()

log_ts_data <- ldata[, -1]
png("acf_plot.png", width = 800, height = 600)
par(mfrow = c(3, 4))
for (i in 1:ncol(log_ts_data)) {
  acf(log_ts_data[[i]], main = paste("ACF of", colnames(log_ts_data)[i]))
}
dev.off()
par(mfrow = c(1, 1))

#Point iii): VAR approach for log series

```

```

log_data_subset <- log_ts_data[, 1:3]
VARF <- function(ts,d){
  var_model <- VAR(ts, p =d , type = "const")
  par(mfrow = c(3, 1))
  for (i in 1:ncol(ts)) {
    acf(resid(var_model)[, i], main = paste("ACF of residuals:", colnames(ts)[i]))
  }
  serial <- serial.test(var_model, lags.pt = 16, type = "PT.asymptotic")
  normality <- normality.test(var_model)
  arch <- arch.test(var_model)
  return(list(var_model,summary(var_model),serial,normality,arch))
}

png("acf_plot_1_1.png", width = 800, height = 600)
VARF(log_data_subset,1)
dev.off()

roots(VARF(log_data_subset,1)[[1]])

',,

',,

png("acf_plot_1_2.png", width = 800, height = 600)
VARF(log_data_subset,2)
dev.off()

#Point iv)
roots(VARF(log_data_subset,1)[[1]])
roots(VARF(log_data_subset,2)[[1]])
#point v)
log_data_subset2 <- log_ts_data[, 10:12]
VARF(log_data_subset2,1)

VARF(log_data_subset2,2)

roots(VARF(log_data_subset2,1)[[1]])
roots(VARF(log_data_subset2,2)[[1]])

#point vi)
VARF(log_ts_data,1)
VARF(log_ts_data,2)

```

```

roots(VARF(log_ts_data,1)[[1]])
roots(VARF(log_ts_data,2)[[1]])
',,
',,
#Exercise 8
log_diff <- log(data[-1, -1] / data[-nrow(data), -1])
date_aligned <- data[-1, 1]

ldata <- cbind(date = date_aligned, log_diff)
log_data_returns <- pivot_longer(as.data.frame(ldata),
                                cols = -date,
                                names_to = "Series",
                                values_to = "LogValue")
png("tsret.png",width = 600,height = 600)
ggplot(log_data_returns, aes(x = date, y = LogValue)) +
  geom_line() +
  facet_wrap(~ Series, scales = "free_y", ncol = 3) +
  labs(title = "Log-Transformed MSCI World and Sector Indices",
       y = "Log(Index)") +
  theme_minimal()
dev.off()
log_ret_data <- ldata[,-1]
png("acfret.png",width = 600,height = 600)
par(mfrow = c(3, 4))

for (i in 1:ncol(log_ret_data)) {
  acf(log_ret_data[[i]], main = paste("ACF of", colnames(log_ret_data)[i]))
}

dev.off()

VARF(log_ret_data[,2:4],2)
VARF(log_ret_data[,10:12],1)
VARF(log_ret_data,1)
VARF(log_ret_data,2)

roots(VARF(log_ret_data,1)[[1]])
roots(VARF(log_ret_data,2)[[1]])

```